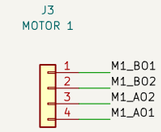
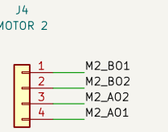
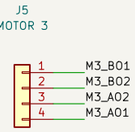


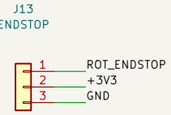
Existing backplane: 12 V, 5 V, UART, I2C0 and dedicated SPI0.

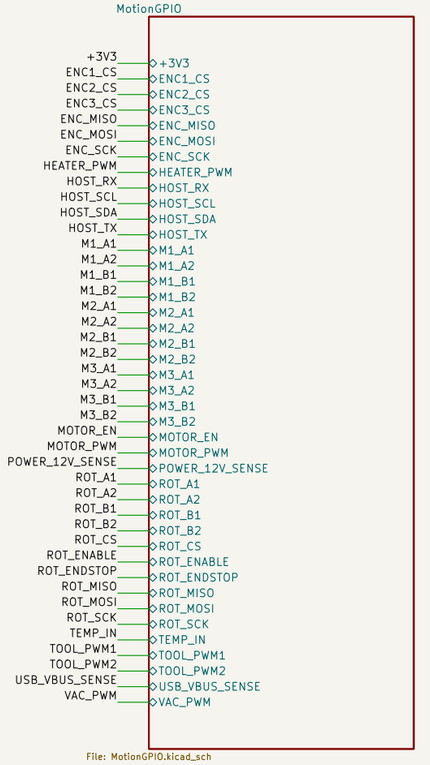
No new pin assignments. All 24 V contacts remain unused.

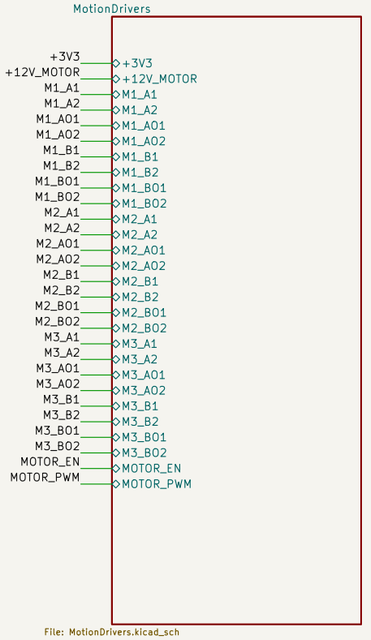








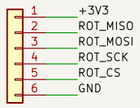


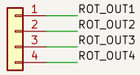


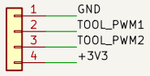
XYZ encoder splitter uses another DIN slot.

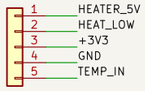
SPI0 is reserved for this motion controller.

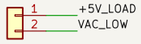
Host GPIO16–21: inputs, no pulls; never drive.

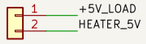


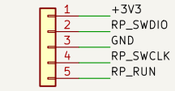


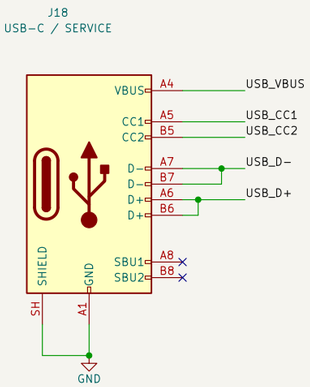


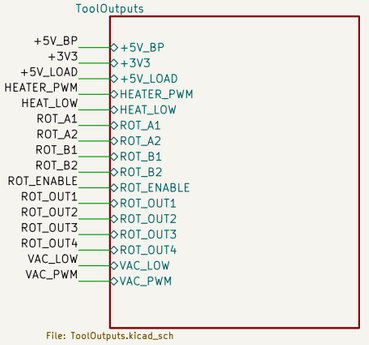












USB: data only; power the backplane before service.

Bare IC daughterboard / engineering review

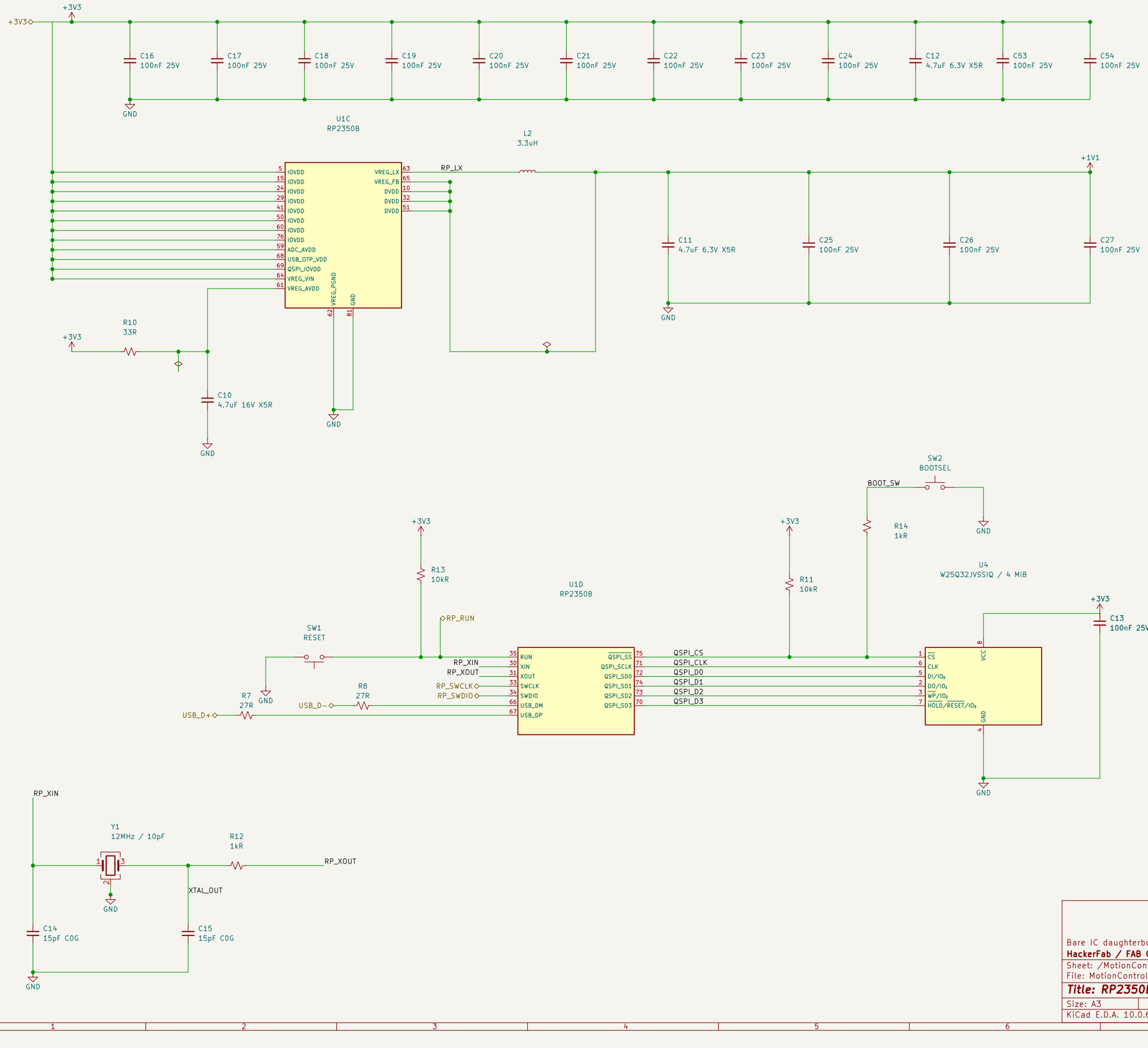
HackerFab / FAB Computer

Sheet: /

File: FabNanoDaughterboard.kicad_sch

Title: FAB Nano

Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/7



Bare IC daughterboard / engineering review

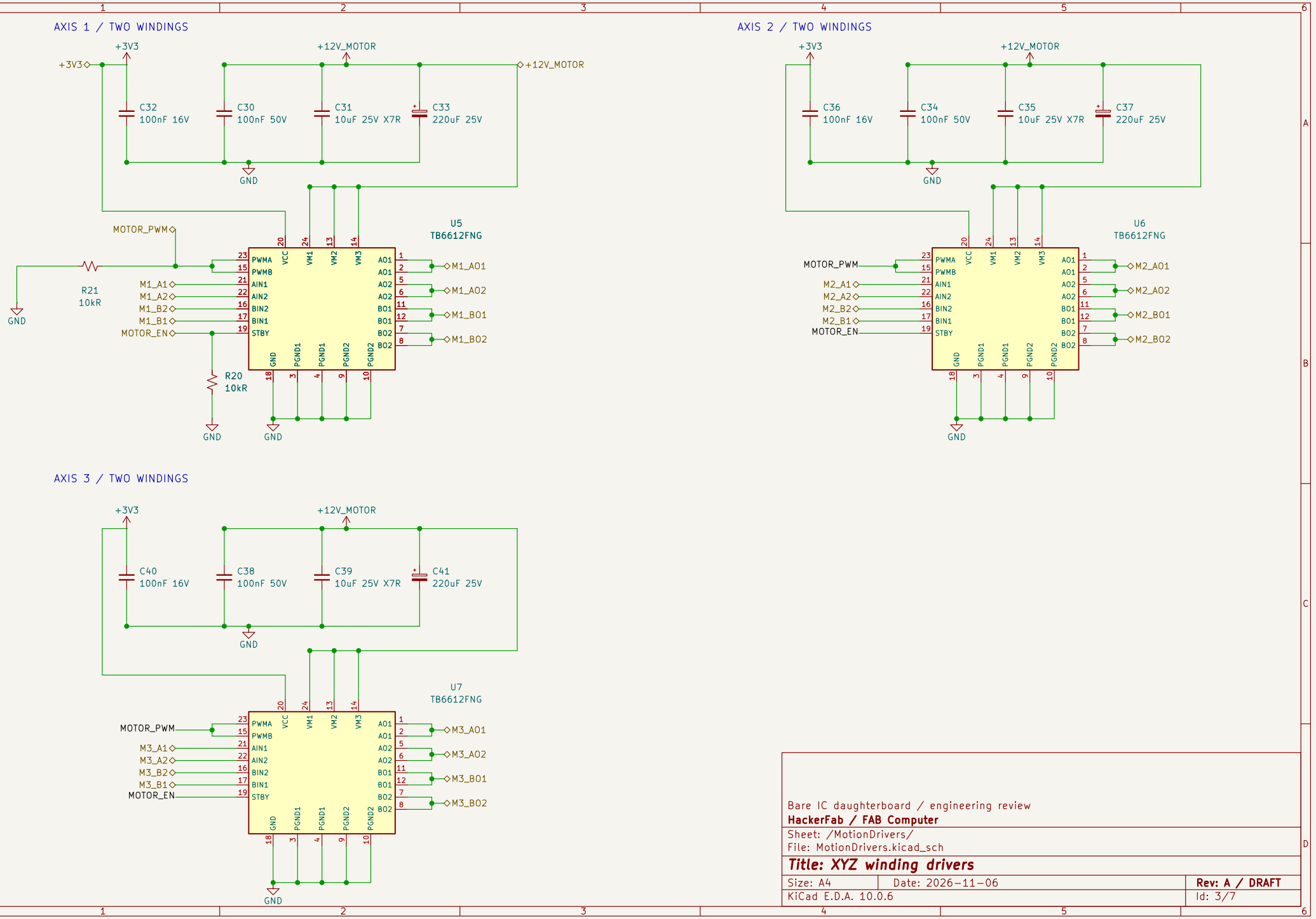
HackerFab / FAB Computer

Sheet: /MotionController/

File: MotionController.kicad_sch

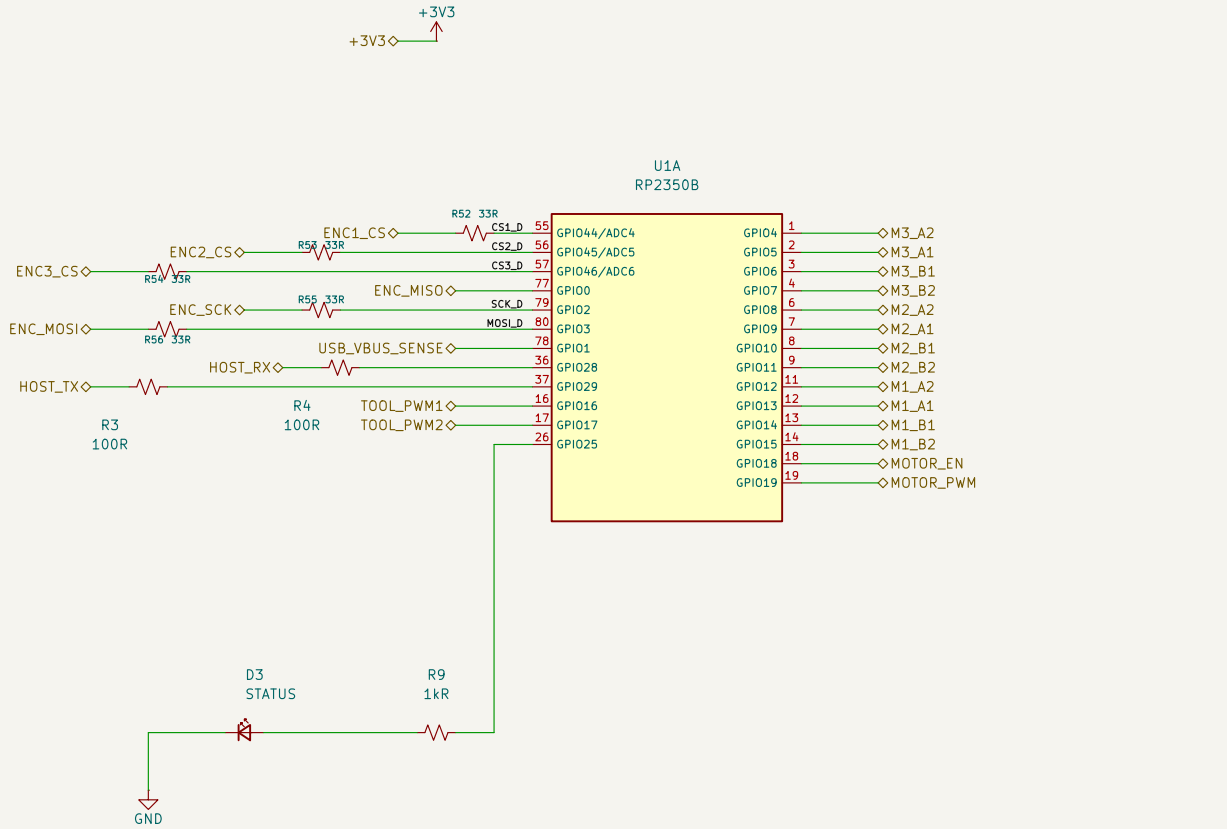
Title: RP2350B

Size: A3	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 2/7

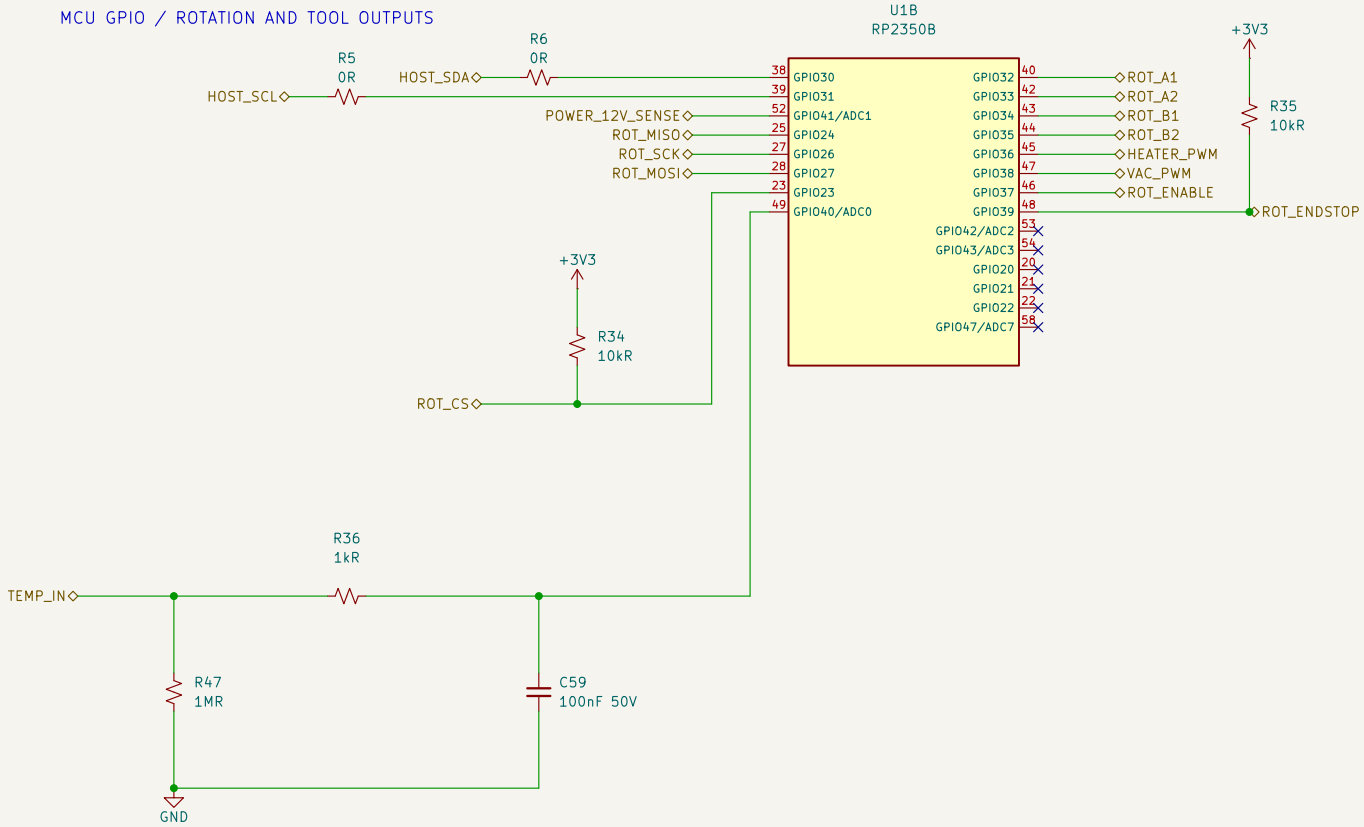

AXIS 1 / TWO WINDINGS
+3V3
+12V_MOTOR
C32
100nF 16V
C30
100nF 50V
C31
10uF 25V X7R
C33
220uF 25V
GND
U5
TB6612FNG
MOTOR_PWM
R21
10kR
M1_A1
M1_A2
M1_B2
M1_B1
MOTOR_EN
R20
10kR
M1_AO1
M1_AO2
M1_BO1
M1_BO2
AXIS 2 / TWO WINDINGS
C36
100nF 16V
C34
100nF 50V
C35
10uF 25V X7R
C37
220uF 25V
U6
TB6612FNG
M2_A1
M2_A2
M2_B2
M2_B1
M2_AO1
M2_AO2
M2_BO1
M2_BO2
AXIS 3 / TWO WINDINGS
C40
100nF 16V
C38
100nF 50V
C39
10uF 25V X7R
C41
220uF 25V
U7
TB6612FNG
M3_A1
M3_A2
M3_B2
M3_B1
M3_AO1
M3_AO2
M3_BO1
M3_BO2
Bare IC daughterboard / engineering review
HackerFab / FAB Computer
Sheet: /MotionDrivers/
File: MotionDrivers.kicad_sch
Title: XYZ winding drivers
Size: A4
Date: 2026-11-06
Rev: A / DRAFT
KiCad E.D.A. 10.0.6
Id: 3/7

MCU GPIO / XYZ MOTION AND ENCODERS

33R at driving pins. Sensor card supplies CS pull-ups.



MCU GPIO / ROTATION AND TOOL OUTPUTS



Bare IC daughterboard / engineering review

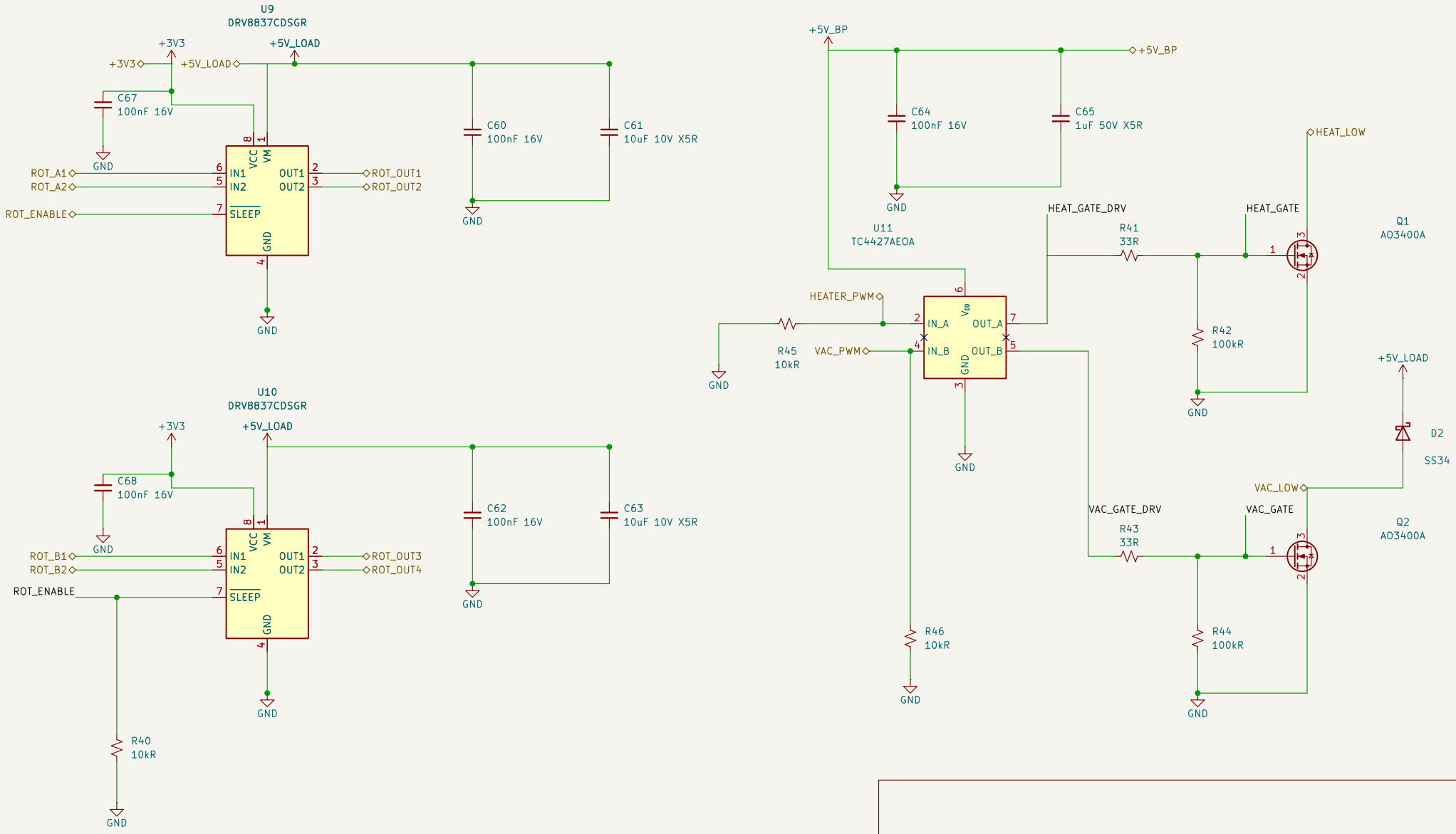
HackerFab / FAB Computer

Sheet: /MotionGPIO/

File: MotionGPIO.kicad_sch

Title: RP2350B / GPIO and interfaces

Size: A4	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 4/7



Thermal cutoff J17 is in series with heater supply.

Headers and load wiring: top-level connector sheet.

Bare IC daughterboard / engineering review

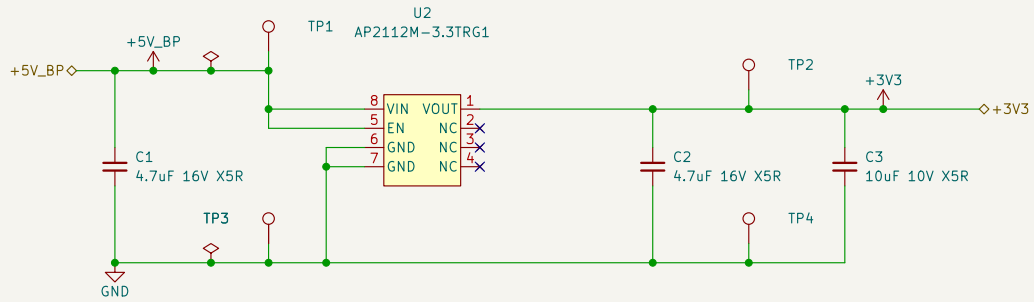
HackerFab / FAB Computer

Sheet: /ToolOutputs/

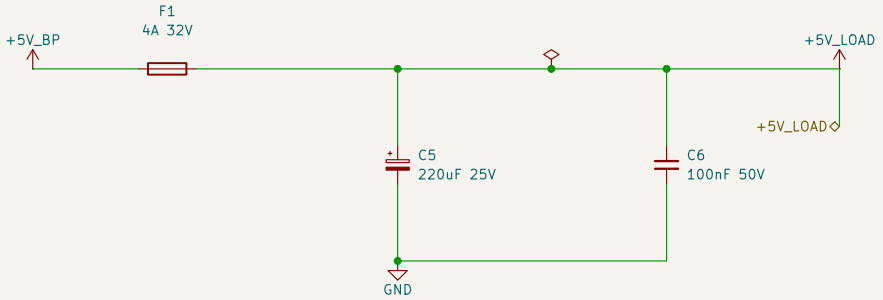
File: ToolOutputs.kicad_sch

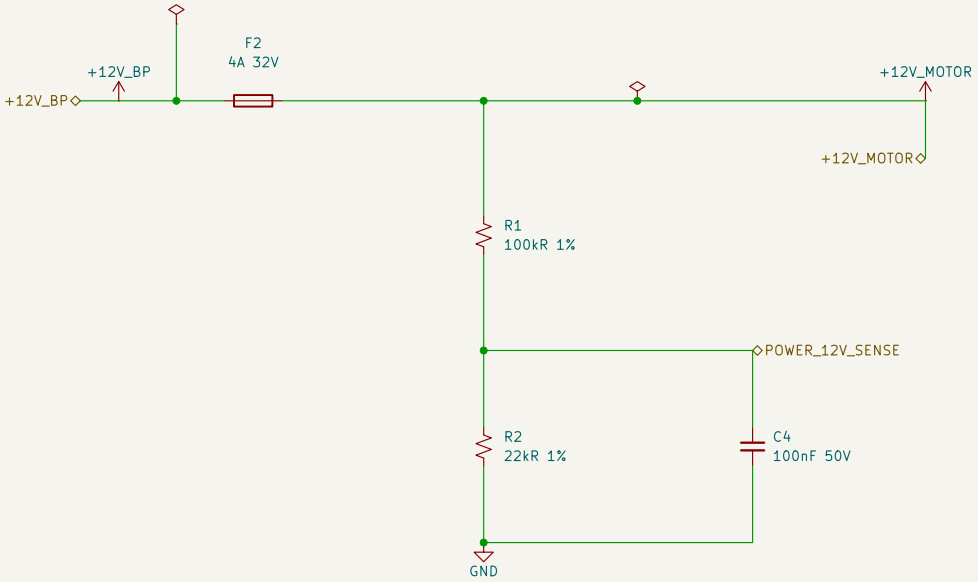
Title: Rotation, heater and vacuum

Size: A4	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 5/7



Logic only. No new backplane contact.





12 V absent: inhibit tool outputs.

Bare IC daughterboard / engineering review

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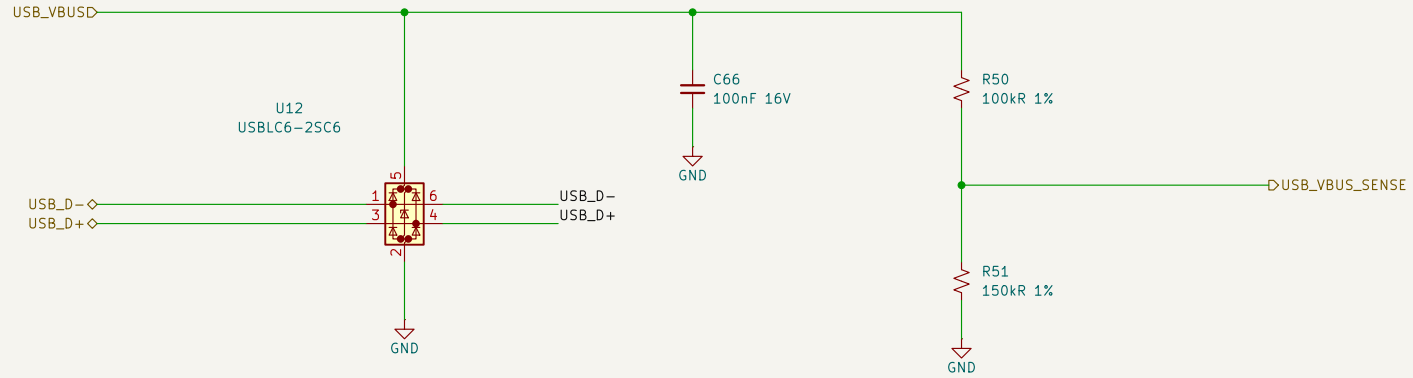
Sheet: /Power/

File: Power.kicad_sch

Title: Backplane rails and interfaces

Size: A4	Date: 2026-11-06	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 6/7

USB SERVICE / ESD PROTECTION AND VBUS DETECTION



USB-C SINK / CC TERMINATIONS



Self-powered USB 2.0 service port. Backplane power must be present.

CC1 and CC2 have independent 5.1k pull-downs. VBUS feeds protection and sensing only.

Bare IC daughterboard / engineering review

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Sheet: /USBInterface/

File: USBInterface.kicad_sch

Title: USB service interface

Size: A4	Date: 2026-11-07	**Rev: A / DRAFT**
KiCad E.D.A. 10.0.6		Id: 7/7